
Analog Electronics Lab

Laboratory Manual

Prepared by: Dr. Dinesh Gopinath

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Course Code	PCEEL308
Semester	S3
Credits	2
Teaching Hours (L:T:P:R)	0:0:3:0
CIE Marks	50
ESE Marks	50
Exam Duration	2 Hrs. 30 Min.
Prerequisites	Nil
Corequisites	PBEET304

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Preface

This manual has been prepared to accompany the laboratory course **Analog Electronics Lab (PCEEL308)(2024 scheme)**, offered in Semester 3 of the B.Tech. programme in Electrical and Electronics Engineering. The course is designed to provide hands-on exposure to the design, simulation, and hardware realisation of fundamental analog electronic circuits employing diodes, BJTs, JFETs, MOSFETs, operational amplifiers, and the 555 timer IC.

Each experiment in this manual follows a uniform structure consisting of the aim, theoretical background, pre-lab assignment, components and equipment required, circuit diagram, design procedure, experimental procedure, expected/model graphs, observation tables, results, and viva-voce questions. Students are expected to complete the pre-lab assignment prior to each laboratory session and to record all readings directly in the laboratory record during the conduct of the experiment.

Course Information

Course Objectives

1. To design transistor and operational-amplifier based electronic circuits.
2. To simulate and implement the designed circuits in hardware and validate experimental results against theoretical/simulated predictions.

Course Outcomes

On successful completion of this course, the student will be able to:

CO	Course Outcome	Bloom's Level
CO1	Use the various electronic instruments for conducting experiments.	K3
CO2	Design and develop various electronic circuits using diodes and Zener diodes.	K3
CO3	Design and implement amplifier and oscillator circuits using BJT and JFET.	K3
CO4	Design and implement basic circuits using IC (OPAMP and 555 timer).	K3
CO5	Simulate electronic circuits using circuit simulation software.	K3
CO6	Use PCB layout software for circuit design.	K3

Table 1: Course Outcomes and Bloom's Knowledge Level

K1: Remember, K2: Understand, K3: Apply, K4: Analyse, K5: Evaluate, K6: Create

Scheme of Evaluation

Continuous Internal Evaluation (CIE – 50 Marks)

Component	Marks
Attendance	5
Preparation / Pre-Lab Work, Viva, and Timely Completion of Lab Reports / Record (Continuous Assessment)	25
Internal Examination	20
Total	50

Table 2: CIE Marks Distribution

The continuous assessment component (25 marks) is further broken down as:

- **Preparation and Pre-Lab Work (7 Marks):** Pre-lab assignments/quizzes and understanding of theoretical background.
- **Conduct of Experiments (7 Marks):** Adherence to procedure, accurate execution, safety, skill proficiency, and teamwork.
- **Lab Reports and Record Keeping (6 Marks):** Clarity, completeness, accuracy, and timely submission.
- **Viva Voce (5 Marks):** Ability to explain the experiment, results, and underlying principles.

The final marks for preparation, conduct, viva, and record are computed as the average across all experiments listed in the syllabus.

End Semester Examination (ESE – 50 Marks)

Component	Marks
Procedure / Preliminary Work / Design / Algorithm	10
Conduct of Experiment / Execution of Work / Troubleshooting / Programming	15
Result with Valid Inference / Quality of Output	10
Viva Voce	10
Record	5
Total	50

Table 3: ESE Marks Distribution

- Students shall be permitted to appear for the end-semester examination only upon submission of a duly certified record.
- The external examiner shall endorse the record at the time of the examination.

CO–PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2	3	3	3	3				3	3		
CO3	2	3	3	3	3				3	3		
CO4	2	3	3	3	3				3	3		
CO5	2	3	3	3	3				3	3		
CO6	3								3	3		

Table 4: CO–PO Mapping (1: Slight, 2: Moderate, 3: Substantial, blank: No Correlation)

Text Books and References

1. Robert T. Paynter, *Introductory Electronic Devices and Circuits*, Pearson Education.
2. Boylestad R. L. and Nashelsky L., *Electronic Devices and Circuit Theory*, Pearson Education.
3. Donald A. Neamen, *Electronic Circuits: Analysis and Design*, McGraw Hill.

General Instructions to Students

Laboratory Conduct

1. Students must bring the laboratory record, observation notebook, graph sheets, and a scientific calculator to every session.
2. The pre-lab assignment for the experiment scheduled for the day must be completed and ready for evaluation before entering the laboratory.
3. Students shall enter the laboratory only with permission and shall occupy the workbench allotted to them.
4. Circuit connections must be checked and approved by the faculty-in-charge or laboratory instructor before the power supply is switched on.
5. Readings must be entered directly into the observation notebook at the time of the experiment; fabricated or estimated readings will not be accepted.
6. Calculations, graphs, and inferences must be completed and the experiment got corrected on the same day, wherever possible, or within the stipulated time announced by the instructor.
7. Students must simulate the circuit using the prescribed circuit simulation software prior to hardware implementation, wherever applicable.
8. Mobile phones shall be kept in silent mode and shall not be used during the conduct of the experiment, except as directed for simulation work.
9. Any damage to equipment or components due to negligence shall be reported immediately and may be subject to the institution's equipment-damage policy.
10. Students are responsible for returning all components, probes, and patch cords to their designated locations and for switching off all instruments and the workbench power supply before leaving the laboratory.

Lab Record Submission

1. The fair record must contain, for each experiment: aim, components/equipment list, circuit diagram, design procedure, experimental procedure, observation table, model graph, calculations, result, and inferences.
2. Records must be certified by the faculty-in-charge within one week of completion of the experiment.
3. Submission of a duly certified record is mandatory for eligibility to appear for the end-semester examination.

Evaluation of Pre-Lab Assignments

Two pre-lab assignments are prescribed by the syllabus prior to the regular experiments:

1. Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a cathode-ray oscilloscope (CRO) / digital storage oscilloscope (DSO).
2. Introduction to circuit simulation using a circuit simulation software package (e.g., LTspice, PSpice, Multisim, or NgSpice/QUCS).

Students are expected to complete these foundational exercises before attempting the numbered experiments listed in [Table 6](#).

Safety Guidelines

Safety Note

1. Always verify the power supply voltage and polarity before connecting it to the circuit. Reversed polarity can permanently damage diodes, transistors, op-amps, and ICs.
2. Switch off the power supply before making, modifying, or removing any circuit connections.
3. Do not exceed the maximum voltage, current, or power ratings of any active or passive component. Always compute and verify current-limiting resistor values before energising a circuit.
4. Use current-limiting resistors in series with LEDs, diodes, and Zener diodes during testing to prevent excessive current flow.
5. Avoid short-circuiting the output terminals of a regulated DC power supply or the output of an operational amplifier.
6. Handle CRO/DSO probes with care; ensure the ground (common) lead of the probe is connected to the circuit ground/common point to avoid ground loops and inaccurate measurements.
7. Be aware of the heat-sink and case temperature of power transistors and voltage regulators during prolonged operation; avoid direct contact.
8. Electrolytic capacitors are polarised. Verify polarity before connecting; reversed polarity may cause the capacitor to overheat, vent, or rupture.
9. Do not work on live circuits with wet hands or in a wet environment.
10. In the event of a component overheating, emitting smoke, or unusual odour, immediately switch off the power supply and inform the laboratory instructor before attempting further investigation.
11. Familiarise yourself with the location of the mains isolation switch and the fire extinguisher in the laboratory.

Precautions

1. Identify the pin configuration (pinout) of every IC and transistor from its datasheet before inserting it into the breadboard.
2. Avoid loose or floating connections, particularly at the input terminals of operational

amplifiers, since floating CMOS/op-amp inputs may result in unpredictable circuit behaviour.

3. Keep signal leads and connecting wires as short as practicable to minimise stray capacitance and inductance, particularly in high-frequency oscillator circuits.
4. Re-verify all connections against the circuit diagram before switching on the power supply, and recheck after every modification.

List of Equipment and Software

Equipment / Software / Components	Typical Use in this Lab
Regulated DC Power Supply (dual/triple output)	Biasing of transistor, op-amp, and 555 timer circuits
Function / Signal Generator	Input excitation for amplifiers, filters, and frequency-response studies
Digital Storage Oscilloscope (DSO)	Measurement of waveforms, amplitude, frequency, and phase shift
Digital Multimeter (DMM)	DC bias point and resistance/voltage/current measurements
Breadboard and connecting wires	Circuit prototyping
Active and Passive Components (diodes, Zener diodes, BJTs, JFETs, MOSFETs, resistors, capacitors, inductors)	Circuit construction across experiments
Operational Amplifier ICs (e.g., μ A741, LM358)	Op-amp based experiments
Comparator IC (e.g., LM311)	Comparator and Schmitt trigger experiments
555 Timer IC	Astable and monostable multivibrator experiments
Circuit Simulation Software (e.g., LTspice / Qspice / NgSpice-QUCS)	Pre-implementation simulation of all designed circuits
PCB Layout Software (e.g., KiCad)	Introduction to PCB design

Table 5: Indicative list of equipment and software used in the course

List of Experiments

Table 6: List of Experiments – Analog Electronics Lab (PCEEL308)

Expt. No.	Title	CO
PL-1	Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a CRO/DSO (Pre-Lab)	CO1
PL-2	Introduction to circuit simulation using a circuit simulation software (Pre-Lab)	CO5
1	Clipping and clamping circuits using diodes	CO2
2	Basic RC circuits – High-pass and Low-pass filters	CO1
3	RC-coupled amplifier using BJT in CE configuration – Measurement of gain, bandwidth, and plotting of frequency response	CO3
4	Emitter follower amplifier	CO3
5	JFET amplifier – Measurement of gain, bandwidth, and plotting of frequency response	CO3
6	MOSFET amplifier	CO3
7	Design and testing of voltage regulators – Zener and series	CO2
8	Design and setup of inverting and non-inverting amplifiers	CO4
9	Op-amp circuits – Scale changer, adder, integrator, and differentiator	CO4
10	Precision rectifier using op-amp	CO4
11	Op-amp oscillators – RC phase-shift and Wien bridge oscillator	CO4
12	Op-amp oscillator – LC oscillator (Colpitts or Hartley)	CO4
13	Waveform generation – Square, triangular, and sawtooth waveform generation using op-amps	CO4
14	Basic comparator and Schmitt trigger circuits using op-amp (using comparator ICs such as LM311)	CO4
15	Active filters (one each of high-pass and low-pass)	CO4
16	Instrumentation amplifier	CO4
17	Astable and monostable multivibrator circuits using 555 timer IC	CO4
18	Introduction to PCB layout software	CO6

Format Followed for Each Experiment

Each experiment in the chapters that follow is presented under the following uniform headings:

1. **Aim**
2. **Pre-Lab Assignment**

3. **Components and Equipment Required**
4. **Theory**
5. **Circuit Diagram**
6. **Design**
7. **Procedure**
8. **Precautions**
9. **Model Graph / Expected Waveform**
10. **Observation Table**
11. **Result**
12. **Viva-Voce Questions**

Chapter 1

Experiment 2: Zener Diode Voltage Regulator

1.1 Aim

To design, and construct a Zener diode based DC voltage regulator for a given load current and input voltage range, and to study its line regulation and load regulation characteristics.

1.2 Apparatus / Components Required

S.No.	Component	Specification
1	Zener diode	$V_Z = 5.6 \text{ V}$, $P_{Z(\text{max})} = 400 \text{ mW}$
2	Resistor R_S	As designed (series/current-limiting resistor)
3	Resistor (as R_L)	1–10 k Ω
4	Regulated DC Power Supply	0–15 V, variable
5	Digital Multimeter	–
6	Breadboard and connecting wires	–

1.3 Theory

A Zener diode, when reverse biased beyond its breakdown voltage V_Z , maintains a nearly constant voltage across its terminals over a wide range of reverse current. This property is exploited to build a simple shunt voltage regulator: a series resistor R_S limits the current drawn from the unregulated input V_{in} , while the Zener diode, connected in parallel with the load R_L , clamps the output voltage to V_Z as long as it is operated between its minimum breakdown-sustaining current $I_{Z(\text{min})}$ and its maximum rated current $I_{Z(\text{max})}$ (set by its power rating).

The regulator must continue to hold $V_{out} \approx V_Z$ under two independent disturbances:

- **Line regulation:** V_{in} varies over its specified range while I_L is held constant.
- **Load regulation:** I_L varies (via R_L) while V_{in} is held constant.

1.4 Design

Given specifications.

$$\begin{aligned}V_Z &= 5.6 \text{ V} \\I_{L(\text{max})} &= 5 \text{ mA} \\V_{in} &: 8 \text{ V to } 15 \text{ V} \quad (V_{in(\text{min})} = 8 \text{ V}, V_{in(\text{max})} = 15 \text{ V}) \\P_{Z(\text{max})} &= 400 \text{ mW}\end{aligned}$$

Step 1: Maximum Zener current from the power rating.

$$I_{Z(\max)} = \frac{P_{Z(\max)}}{V_Z} = \frac{400 \text{ mW}}{5.6 \text{ V}} \approx 71.4 \text{ mA}$$

Step 2: Choose minimum Zener current. To ensure the diode stays in the breakdown region even at worst case (minimum V_{in} , maximum I_L), a minimum holding current is chosen as

$$I_{Z(\min)} = 5 \text{ mA} \quad (\text{a conservative, commonly used design margin})$$

Step 3: Design R_S for the worst case (minimum V_{in} , maximum I_L). At $V_{in(\min)}$, the source must supply both $I_{L(\max)}$ and $I_{Z(\min)}$:

$$R_S = \frac{V_{in(\min)} - V_Z}{I_{L(\max)} + I_{Z(\min)}} = \frac{8 - 5.6}{(5 + 5) \times 10^{-3}} = \frac{2.4}{0.010} = 240 \, \Omega$$

Choosing the nearest standard (E12) value: $R_S = 220 \, \Omega$.

Step 4: Verify at minimum V_{in} , full load (checks regulation is maintained).

$$I_T = \frac{V_{in(\min)} - V_Z}{R_S} = \frac{2.4}{220} \approx 10.9 \text{ mA}$$

$$I_Z = I_T - I_{L(\max)} = 10.9 - 5 = 5.9 \text{ mA} \quad (> I_{Z(\min)} = 5 \text{ mA}) \quad \checkmark$$

Step 5: Verify at maximum V_{in} , no load (checks Zener power rating is not exceeded).

$$I_Z = \frac{V_{in(\max)} - V_Z}{R_S} = \frac{15 - 5.6}{220} \approx 42.7 \text{ mA}$$

$$P_Z = I_Z \times V_Z = 42.7 \text{ mA} \times 5.6 \text{ V} \approx 239 \text{ mW} \quad (< 400 \text{ mW}) \quad \checkmark$$

Step 6: Power rating of R_S .

$$P_{R_S} = \frac{(V_{in(\max)} - V_Z)^2}{R_S} = \frac{(9.4)^2}{220} \approx 0.40 \text{ W}$$

Use a $220 \, \Omega$, 1 W resistor (adequate safety margin over the calculated 0.40 W).

Final design summary.

Parameter	Value
Zener diode	5.6 V, 400 mW
Series resistor R_S	220 Ω , 1 W
Load current range	0 to 5 mA
Input voltage range	8 V to 15 V
Regulated output	$\approx 5.6 \text{ V}$

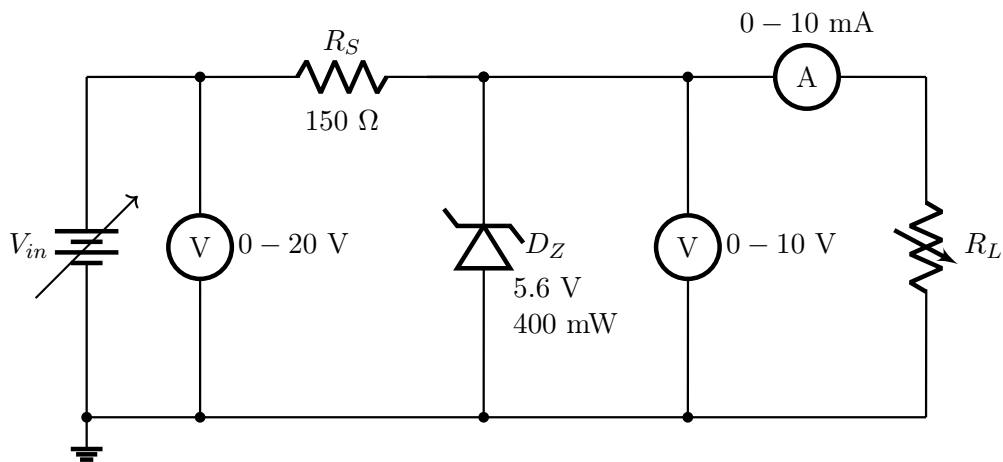


Figure 1.1: Zener diode voltage regulator circuit

1.5 Circuit Diagram

1.6 Procedure

(a) Line regulation.

1. Connect the circuit as shown in Figure 1.1 with R_L set to give $I_L = I_{L(\max)} = 5 \text{ mA}$ (i.e., $R_L = V_Z / I_{L(\max)} \approx 1.12 \text{ k}\Omega$).
2. Vary V_{in} in steps from 8 V to 15 V, keeping R_L fixed.
3. Note V_{out} across the Zener/load for each value of V_{in} .
4. Plot V_{out} versus V_{in} .

(b) Load regulation.

1. Fix V_{in} at a convenient value within the design range (e.g., $V_{in} = 12 \text{ V}$).
2. Vary R_L (using the decade resistance box) so that I_L varies from 0 (open circuit) to $I_{L(\max)} = 5 \text{ mA}$.
3. Note V_{out} for each value of I_L .
4. Plot V_{out} versus I_L .

1.7 Tabulation

Line regulation (I_L held at 5 mA).

V_{in} (V)	V_{out} (V)
8	
9	
10	
11	
12	
13	
14	
15	

Load regulation (V_{in} held constant).

I_L (mA)	V_{out} (V)
0	
1	
2	
3	
4	
5	

1.8 Model Graph

Plot two graphs from the observations:

1. V_{out} (Y-axis) vs. V_{in} (X-axis) — should be nearly flat once V_{in} exceeds the value needed to bring the Zener into breakdown, confirming line regulation.
2. V_{out} (Y-axis) vs. I_L (X-axis) — should remain close to V_Z up to $I_{L(max)}$, with a slight droop as I_L approaches the design limit, confirming load regulation.

1.9 Calculations from Observations

$$\% \text{ Line regulation} = \frac{\Delta V_{out}}{\Delta V_{in}} \times 100\%$$

$$\% \text{ Load regulation} = \frac{V_{out(\text{no load})} - V_{out(\text{full load})}}{V_{out(\text{full load})}} \times 100\%$$

1.10 Result

A Zener diode voltage regulator was designed for $V_Z = 5.6$ V, $I_{L(max)} = 5$ mA, over an input range of 8–15 V, using a 400 mW Zener diode and a series resistor $R_S = 220 \Omega$. The circuit was constructed and its line regulation and load regulation characteristics were verified experimentally.

1.11 Questions

1. Why is the Zener diode operated in reverse breakdown for voltage regulation?
2. What determines the maximum and minimum permissible Zener current?
3. Why must R_S be designed using the minimum input voltage and maximum load current condition?
4. What happens to the regulator if R_L is accidentally open-circuited?
5. Define line regulation and load regulation. What is an ideal value for each?
6. How does the power rating of the Zener diode limit the input voltage range for a fixed R_S ?